Viscosity:

$$F_{viscous} = \mu A \frac{\Delta v_x}{\Delta y}$$

Injection moulding cavity:

$$D_c = D_p(1+S+S^2)$$

Mass continuity:

$$Q = Av = \text{constant}$$

Bernoulli's:

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + h_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + h_2 + f$$

In metal casting:

$$v = \sqrt{2gh}$$

Reynolds number:

$$Re = \frac{\rho v d}{\mu}$$

Chvorinov's rule:

$$T_{TS} = C_m\left(\frac{V}{A}\right)^n$$

$$n = 2$$

Drawing force:

$$F = \pi D_p t (TS)\left(\frac{D_b}{D_p} - 0.7\right)$$

Blank holding force:

$$F_h = 0.015 Y \pi \{D_b^2 - (D_p + 2.2t + 2R_d)^2\}$$

Clearance: $c = \alpha t$

Maximum blanking force:

$$F = STL = 0.7(UTS)TL$$

Bend allowance:

$$L_b = \alpha(R + kT)$$

$$k = \begin{cases} 0.33 & R < 2T \\ 0.5 & R \geq 2T \end{cases}$$

Bending strain:

$$\varepsilon = \frac{1}{\frac{2R}{T} + 1}$$

Spring back:

$$\frac{R_i}{R_f} = 4\left(\frac{R_i Y}{ET}\right)^3 - 3\left(\frac{R_i Y}{ET}\right) + 1$$

Bending forces:

V-die	Wipe bending
$F = \frac{1.33(UTS)LT^2}{W}$	$F = \frac{0.33(UTS)LT^2}{W}$

Drawing ratio:

$$DR = \frac{D_b}{D_p} \leq 2.0$$

Reduction:

$$r = \frac{D_b - D_p}{D_b} < 0.50$$

Thickness to diameter ratio:

$$\frac{t}{D_b} > 0.01$$

Cast iron with high carbon content and gray iron have NO SHRINKAGE.